

ANALYSIS_STATUS: SUCCEEDED

Computational Replication Report —

Rosacea Hypothesis

etr_lipid_metabolism_acsl5_macrophage

Hypothesis label: Lipid-metabolic reprogramming centred on ACSL5 drives M1 macrophage infiltration in erythematotelangiectatic rosacea (ETR).

Dataset: `geo:GSE65914` (platform GPL570, Affymetrix HG-U133 Plus 2.0).

Prespecified genes: ACSL5, ACADVL, PPARG, CPT1A, CD68, CD163, CD86.

Comparison executed: ETR (n = 14 arrays) vs. healthy control (n = 20 arrays), per-gene Welch two-sample t-test with Benjamini–Hochberg (BH) FDR across the 7 genes, plus within-ETR Spearman correlation of ACSL5 against each macrophage marker.

Summary

This was a computational replication, not a literature review: the GSE65914 series matrix and the GPL570 annotation were downloaded from the NCBI GEO FTP mirror, samples were classified, probes were mapped and aggregated to gene level, and the prespecified statistical contrast was executed and locked as reproducible artifacts. A clean replay of `analysis.py` into a separate directory produced **byte-identical** `samples.csv`, `gene_results.csv`, and `spearman_acsl5_macrophage.csv`, confirming determinism. The `MANIFEST.yaml` bundle is closed with `status: SUCCEEDED`, `fallback_used: false`, and `direct_analysis_completed: true`.

The core observations replicate cleanly and are quantitatively strong. In the ETR-versus-control contrast, the lipid-metabolism hub gene **ACSL5** is the single most strongly up-regulated target (fold change $\approx 5.6\times$, BH $q = 3.3 \times 10^{-13}$, Cohen's $d = 4.23$), accompanied by the PPAR master regulator **PPARG** ($\approx 4.5\times$) and the very-long-chain fatty-acid dehydrogenase **ACADVL** ($\approx 2.0\times$). All three macrophage markers — **CD68**, **CD163**, and **CD86** — are significantly elevated, consistent with increased macrophage abundance in lesional ETR skin. The fatty-acid-oxidation gene **CPT1A** is essentially unchanged (fold change ≈ 1.0 , $q = 0.97$), serving as an internal negative control that argues against a blanket up-regulation of the entire fatty-acid-oxidation program. These conclusions survive a subject-level, pseudoreplication-controlled sensitivity analysis in which the two replicate arrays per subject are averaged.

However, the *mechanistic* claim in the hypothesis label — that ACSL5-centred lipid reprogramming *drives M1 macrophage infiltration* — is **not demonstrated** by the data. Within ETR samples, ACSL5 shows no significant correlation with any macrophage marker (all Spearman $\rho > 0.45$), so there is no expression-level coupling to support a driver relationship. Moreover, the macrophage signal shows **no M1-polarization skew**: the relative CD86(M1)/CD163(M2) balance shifts *toward* the M2 marker CD163 in ETR ($p = 0.018$, Cohen's $d = -0.85$), and the M2-polarization master regulator PPARG is strongly up-regulated. The correct interpretation is therefore a **partial replication**: lipid-metabolic reprogramming and macrophage infiltration clearly co-occur in ETR, but the specific ACSL5-driven-M1 axis is not supported by these transcriptomic data.

Key Findings

Finding F001 — ACSL5, ACADVL and PPARG are strongly up-regulated alongside macrophage markers in ETR, while CPT1A is null and ACSL5 is uncorrelated with macrophage markers within ETR

The prespecified contrast was ETR ($n = 14$ arrays) versus healthy control ($n = 20$ arrays) in GSE65914 on GPL570, using a per-gene Welch two-sample t-test on gene-level $\log_2$ expression, with BH-FDR applied across the 7 prespecified genes. The GEO-supplied values were already normalized on the $\log_2$ scale and were used as provided (no re-transformation). Gene-level expression was computed as the arithmetic mean of curated-annotation probes mapping to each gene. Cohen's d was oriented ETR-minus-control with a pooled standard deviation; fold change was computed as $2^{(\log_2 \text{ mean difference})}$.

The six-gene up-regulation signal is unambiguous and the effect sizes are exceptionally large. **ACSL5** is the standout, rising 5.62-fold with a BH q -value of 3.31×10^{-13} and a Cohen's d of 4.23 — an effect size far beyond conventional "large" thresholds. **PPARG** rises 4.48-fold ($q = 8.37 \times 10^{-10}$, $d = 2.93$) and **ACADVL** 2.03-fold ($q = 4.04 \times 10^{-9}$, $d = 3.92$). The three macrophage markers all reach significance: **CD86** 1.40-fold ($q = 5.05 \times 10^{-5}$, $d = 2.20$), **CD68** 1.60-fold ($q = 2.14 \times 10^{-4}$, $d = 1.79$), and **CD163** 1.76-fold ($q = 3.33 \times 10^{-4}$, $d = 1.65$). In sharp contrast, **CPT1A** is flat (fold change 1.00, $q = 0.97$, $d = -0.01$), providing a clean internal negative control within the same panel.

Gene	Role in hypothesis	Fold change (ETR/control)	BH q-value	Cohen's d	Direction
ACSL5	Lipid-metabolism hub	5.62	3.31×10^{-13}	4.23	▲ Up
PPARG	PPAR / fatty-acid, M2 regulator	4.48	8.37×10^{-10}	2.93	▲ Up
ACADVL	Very-long-chain FA dehydrogenase	2.03	4.04×10^{-9}	3.92	▲ Up
CD86	M1 macrophage marker	1.40	5.05×10^{-5}	2.20	▲ Up
CD68	Pan-macrophage marker	1.60	2.14×10^{-4}	1.79	▲ Up
CD163	M2 macrophage marker	1.76	3.33×10^{-4}	1.65	▲ Up
CPT1A	Fatty-acid oxidation	1.00	0.97	-0.01	— Unchanged

The second, mechanistic half of the hypothesis was tested by within-ETR Spearman correlations of ACSL5 against each macrophage marker (n = 14 arrays). None reached significance: ACSL5–CD68 rho = 0.30 (p = 0.30), ACSL5–CD163 rho = 0.04 (p = 0.90), and ACSL5–CD86 rho = 0.31 (p = 0.28), all with BH q > 0.45. In other words, the samples that most strongly up-regulate ACSL5 are *not* preferentially the samples with the highest macrophage marker expression. Group-level co-elevation exists (both lipid genes and macrophage markers are higher in ETR than control), but this reflects the shared disease state rather than a within-disease driver relationship. Cross-sectional expression data of this kind cannot in any case establish causal direction.

Finding F002 — Subject-level (pseudoreplication-controlled) reanalysis confirms the same pattern

A key structural feature of GSE65914 is that each subject contributed two replicate arrays (suffixes `_1/``_2`). The prespecified contrast treated arrays as independent observations, which risks anticonservative p-values through pseudoreplication. As a sensitivity analysis, the two replicate arrays per subject were averaged, collapsing the data to ETR $n = 7$ subjects versus control $n = 10$ subjects, and the contrast was re-run.

The subject-level analysis reproduces the array-level result almost exactly, with identical fold changes and near-identical effect sizes: ACSL5 fold change 5.62 ($q = 4.4 \times 10^{-5}$, $d = 4.19$); ACADVL 2.03 ($q = 1.1 \times 10^{-4}$, $d = 3.88$); PPARG 4.48 ($q = 1.1 \times 10^{-4}$, $d = 2.88$); CD86 1.40 ($q = 8.7 \times 10^{-3}$, $d = 2.23$); CD68 1.60 ($q = 1.4 \times 10^{-2}$, $d = 1.95$); CD163 1.76 ($q = 1.7 \times 10^{-2}$, $d = 1.60$); and CPT1A flat at 1.00 ($q = 0.98$, $d = -0.02$). The within-ETR Spearman correlations remained non-significant at the subject level ($n = 7$): ACSL5–CD68 $\rho = 0.29$ ($p = 0.53$), ACSL5–CD163 $\rho = 0.07$ ($p = 0.88$), ACSL5–CD86 $\rho = 0.43$ ($p = 0.34$). This demonstrates that the up-regulation findings are not an artifact of counting each subject twice, and that the absence of ACSL5–macrophage coupling is likewise robust to the aggregation choice. These diagnostic tables were saved under `sensitivity/` (outside the locked artifact bundle).

This robustness matters for interpretation. Because doubling the sample by replicate arrays would tend to *inflate* apparent significance, one might worry the primary up-regulation p-values are too optimistic. Collapsing to genuine biological replicates removes that concern — the six positive genes remain significant after multiple-testing correction (all $q < 0.02$), and the negative results (CPT1A; ACSL5–macrophage correlations) remain negative. The picture is internally consistent across both analysis granularities.

Finding F003 — Macrophage up-regulation reflects abundance, not M1 skew; the relative balance leans M2

Because the hypothesis specifically invokes *M1* macrophage infiltration, the M1/M2 balance was examined directly rather than inferred from any single marker. All three markers rise in ETR (CD68 1.60 $\times$, CD86 1.40 $\times$, CD163 1.76 $\times$), a pattern consistent with an overall increase in macrophage abundance. But the *relative* polarization does not favour M1. A log2 CD86(M1)-minus-CD163(M2) skew index is significantly *lower* in ETR than control (mean -0.768 vs. -0.434 ; difference -0.334 ; Welch $t = -2.50$; $p = 0.018$; Cohen's $d = -0.85$), i.e. a shift toward the M2 marker CD163. CD86 relative to the pan-macrophage marker CD68 trends downward (CD86–CD68 difference -0.194 ; $p = 0.067$; $d = -0.66$), while CD163 relative to CD68 is

essentially unchanged (difference +0.140; $p = 0.28$). Reinforcing this, the M2-polarization master regulator **PPARG** is one of the most strongly up-regulated genes in the entire panel (fold change 4.48; $q = 8.4 \times 10^{-10}$). Taken together, the transcriptomic signature is best described as increased macrophage infiltration *without* an M1 bias, and if anything with a relative M2/PPARG lean. The diagnostic table is saved as `sensitivity/m1_m2_skew.csv`.

The caveat is that bulk-tissue marker expression cannot definitively resolve macrophage polarization state — CD86, CD163, and CD68 are expressed by other cell types and the ratios are population averages. Nevertheless, three converging indices (CD86–CD163 skew, CD86–CD68 trend, and PPARG up-regulation) all point away from an M1-dominant program, so the conclusion is directionally well supported even if the absolute polarization fractions are unknown.

Mechanistic Model / Interpretation

The data cleanly separate into an **observed** layer that replicates and an **inferred mechanistic** layer that does not.

OBSERVED (replicated, strong effect sizes)	MECHANISTIC CLAIM (NOT supported)
ETR lesional skin vs. healthy control:	
ACSL5 ▲▲▲ (5.6x, d=4.23)	Hypothesis: ACSL5 lipid reprogramming DRIVES M1 macrophage infiltration
PPARG ▲▲ (4.5x)	
ACADVL ▲ (2.0x)	
CPT1A – (unchanged)	
lipid-metabolic reprogramming	
CD68 ▲ (1.6x)	requires: ▼
CD163 ▲ (1.8x)	
CD86 ▲ (1.4x)	
macrophage infiltration	
(a) within-ETR ACSL5 <-> macrophage coupling --> NOT FOUND (q>0.45)	
(b) M1 polarization skew --> NOT FOUND (leans M2, p=0.018)	
(c) causal direction --> cannot be tested (cross-sectional)	
M1/M2 skew --> leans toward M2 (CD163), PPARG (M2 regulator) up	

What is solidly established. Lipid-metabolic reprogramming and macrophage infiltration co-occur in erythematotelangiectatic rosacea. ACSL5 is the strongest single signal in the panel — a genuinely striking finding given its role as a fatty-acid-activating hub. The co-elevation of PPARG and ACADVL, with CPT1A unchanged, points to a *selective* remodeling of lipid handling

(acyl-CoA synthesis and very-long-chain fatty-acid dehydrogenation up; mitochondrial fatty-acid-oxidation entry via CPT1A unchanged) rather than a uniform up- or down-shift of the whole pathway.

What is not established. Three separate lines of evidence undercut the specific "ACSL5 → M1" mechanism. (1) There is no within-ETR correlation between ACSL5 and any macrophage marker, so the two co-elevated modules are not linked at the sample level. (2) The macrophage compartment is not M1-skewed; the relative balance leans toward the M2 marker CD163, and the M2 regulator PPARG is one of the most up-regulated genes. (3) Even the group-level co-elevation is cross-sectional and cannot assign causal direction. The hypothesis is therefore **partially supported**: its descriptive premises (lipid reprogramming; macrophage infiltration) hold, but its causal-directional and polarization-specific claims do not.

Evidence Base

The replication is anchored in three papers whose abstracts were reviewed and whose relevant passages were verified against the stored text.

PMID	Title	Role in this analysis
PMID: 25848978	<i>Molecular and Morphological Characterization of Inflammatory Infiltrate in Rosacea Reveals Activation of Th1/Th17 Pathways</i>	Source/context study for the macrophage-infiltration pillar
PMID: 36831330	<i>Sirtuin 6 — A Key Regulator of Hepatic Lipid Metabolism and Liver Health</i>	Establishes ACSL5 as a lipid/FAO hub and links lipid metabolism to M1↔M2 polarization
PMID: 32347412	<i>Roles of ACSL5 and CSF2 in inhibition of palmitic or stearic acids in lung cancer cell proliferation and metabolism</i>	Independent evidence that ACSL5 is a lipid-responsive metabolic gene

Macrophage infiltration in ETR. [PMID: 25848978](#) directly supports the observed macrophage-marker up-regulation, characterizing "erythematotelangiectatic rosacea (ETR) already as a disease with significant influx of proinflammatory cells," and noting that "Macrophages and mast cells are increased in all three subtypes of rosacea." These statements corroborate the elevated CD68, CD163, and CD86 signal (Findings F001/F003). Notably, this study frames the

infiltrate around Th1/Th17 activation rather than a specifically M1-polarized macrophage program, which is consistent with our observation that the macrophage compartment is not M1-skewed.

ACSL5 as a lipid hub linked to macrophage polarization. PMID: 36831330 provides the mechanistic rationale for treating ACSL5 as the lipid-metabolic centre of the hypothesis, stating that "SIRT6 can also directly modulate acyl-CoA synthetase long chain family member 5 (ACSL5) activity for fatty acid oxidation," and that such lipid-metabolic regulation can "dampen hepatic inflammation through the modulation of macrophage polarization from M1 to M2 type." This is the conceptual bridge the hypothesis relies upon — but it is worth emphasizing that in the cited biology the direction is *toward* M2, which aligns with our finding that ETR leans M2/PPARG rather than M1.

ACSL5 as a lipid-responsive gene. PMID: 32347412 independently identifies ACSL5 as a metabolism-associated gene whose expression is up-regulated by specific fatty acids (palmitic/stearic acid) in a time- and dose-dependent manner, reinforcing the plausibility of ACSL5 as a node that responds to and remodels lipid metabolism. This supports the biological credibility of the strong ACSL5 up-regulation observed here, without speaking to its causal role in macrophage recruitment in skin.

Collectively, the literature *supports* the two descriptive pillars (lipid reprogramming and macrophage infiltration) and even the general lipid→polarization concept, but — read carefully — it points toward an M2 rather than M1 outcome, which matches our data and argues against the specific "M1" wording of the hypothesis.

Limitations and Knowledge Gaps

- **Pseudoreplication.** Each subject contributed two arrays (1/2). The prespecified contrast treated arrays as independent, so nominal p-values in the primary analysis may be anticonservative. The subject-level sensitivity analysis (Finding F002) mitigates this and reproduces every conclusion, but the primary `gene_results.csv` retains the array-level counts by design.
- **Underpowered correlations.** The within-ETR Spearman analysis rests on 14 arrays (7 subjects × 2) and, at the subject level, only 7 subjects. Non-significant correlations cannot exclude a modest true ACSL5–macrophage association; absence of evidence is not evidence of absence.

- **Causality.** These are cross-sectional expression data. Even the robust group-level co-elevation of lipid genes and macrophage markers cannot establish whether ACSL5 activity drives macrophage infiltration, the reverse, or whether both are downstream of a shared upstream trigger.
- **Probe-to-gene aggregation.** Gene-level values are arithmetic means of curated-annotation probes on the log2 scale. In particular, CD68 is represented only by the multi-gene probe `203507_at` (annotated LOC101928634//SNORA67//CD68), so its signal is not cleanly CD68-specific and should be interpreted with caution.
- **Marker-based polarization.** M1/M2 status is inferred from a small marker set (CD86, CD163, CD68, PPARG) rather than a full deconvolution or single-cell readout. The M2-lean conclusion is directionally consistent across multiple indices but remains a bulk-tissue inference.
- **Source normalization.** GEO-supplied normalized log2 values were used as provided; the analysis did not reprocess raw CEL files, so any platform-level normalization idiosyncrasies are inherited.

Proposed Follow-up Experiments / Actions

1. **Single-cell or spatial transcriptomics of ETR lesional skin** to resolve whether ACSL5 is expressed in macrophages themselves or in keratinocytes/endothelium, and to measure per-cell co-expression of ACSL5 with M1 vs. M2 programs — the only way to properly test a within-tissue driver relationship that bulk correlation (n = 14) cannot.
2. **Larger, replicate-independent ETR cohort** with one array/library per subject to power the within-disease ACSL5–macrophage correlation and to formally test the M1/M2 skew with adequate sample size.
3. **Explicit M1/M2 deconvolution** (e.g., CIBERSORTx or a curated polarization signature) on the existing data and on new cohorts to quantify polarization state beyond individual markers, confirming or refuting the M2-lean signal.
4. **Functional perturbation** — ACSL5 knockdown/overexpression in relevant skin macrophage or keratinocyte models, followed by macrophage chemotaxis and polarization assays — to test causal direction directly rather than by association.

5. **Lipidomic profiling** of ETR skin to determine whether the ACSL5/ACADVL up-regulation (with unchanged CPT1A) corresponds to accumulation of specific acyl-CoA or very-long-chain fatty-acid species, connecting the transcriptomic signal to actual lipid flux.
6. **Cross-subtype comparison** (ETR vs. papulopustular vs. phymatous rosacea, both excluded here) to test whether the ACSL5/lipid signature is ETR-specific or shared across the rosacea spectrum.

Artifacts (locked, replay-verified)

All artifacts live under `kb/hypotheses/Rosacea/etr_lipid_metabolism_acsl5_macrophage/openscientist_artifacts/`:

- `MANIFEST.yaml` — `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`; records input accessions (GSE65914 series matrix; GPL570 annotation) with credential-free NCBI FTP URLs, byte counts, and SHA-256 checksums; all output paths/roles/checksums; the exact comparison; and limitations.
- `analysis.py` (role CODE) — standalone deterministic analysis accepting `--output-dir` and `--cache-dir`.
- `environment.txt` (role ENVIRONMENT) — Python version, OS, and exact package versions.
- `methods.md` (role METHODS) and `comparison.md` (role SUMMARY).
- `samples.csv`, `gene_results.csv`, `spearman_acsl5_macrophage.csv` (role TABULAR_RESULT); `probe_gene_map.csv` (role INPUT_MANIFEST).
- `raw/` — recoverable downloaded inputs; `replay/` — clean re-execution outputs verified **byte-identical** to the primary tabular results (`byte_identity: true` for `samples.csv`, `gene_results.csv`, and `spearman_acsl5_macrophage.csv`).

The clean replay command `python3 analysis.py --output-dir ./replay --cache-dir ./raw` was executed and `verified: true`. Primary outputs and checksums were locked before any cross-provider comparison; no external provider results influenced method, sample, feature, probe, model, threshold, or parameter choices.

Conclusion

The replication **SUCCEEDED** and **partially supports** the hypothesis. ACSL5 is the most strongly up-regulated target gene in ETR (5.6×, $q = 3.3 \times 10^{-13}$), with PPARG and ACADVL also elevated and CPT1A unchanged, while all three macrophage markers are significantly increased — results robust to subject-level reanalysis. But the specific mechanistic claim fails: within ETR, ACSL5 does not correlate with any macrophage marker (all $q > 0.45$), and the macrophage compartment shows no M1 skew — the relative balance leans toward the M2 marker CD163 ($p = 0.018$) with strong PPARG up-regulation. Lipid-metabolic reprogramming and macrophage infiltration clearly co-occur in ETR, but the data do not demonstrate that ACSL5-centred lipid reprogramming *drives M1* macrophage infiltration.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery